

Step 5 Validation Protocol: Temperature and Field Scans

1 Purpose of the test

Step 4 showed that discrete temperature and field controls can separate a lattice-driven channel from a finite local spin-correlation control. Step 5 turns that logic into continuous scans. The test asks whether finite-window spectral observables follow the phenomenological spin-Peierls order parameter $\delta(T, B)$ and its associated dynamical lattice coupling g_Q^{eff} , rather than merely reflecting the presence of a finite $C_1(T, B)$.

2 Physical model

The Hamiltonian is unchanged from Step 4:

$$H_{\text{mix}} = V_{BD}^{\text{static}} L_{BD} + g_Q^{\text{eff}} Q L_{BD}, \quad (1)$$

with

$$V_{BD}^{\text{static}} = V_0 + \lambda_\delta \delta(T, B) + \lambda_C C_1(T, B). \quad (2)$$

For the lattice scans,

$$g_Q^{\text{eff}} = g_{Q0} \frac{\delta(T, B)}{\delta(T_{\text{ref}}, B_{\text{ref}})}. \quad (3)$$

For the C_1 static scan, $g_Q^{\text{eff}} = 0$ and the scalar spin-correlation channel is kept active.

3 Trend definitions

The spin-Peierls transition temperature is

$$T_{\text{SP}}(B) = \max [0, T_{\text{SP},0}(1 - \alpha_B B^2)]. \quad (4)$$

The dimerisation follows

$$\delta(T, B) = \begin{cases} \delta_0 (1 - T/T_{\text{SP}}(B))^\beta, & T < T_{\text{SP}}(B), \\ 0, & T \geq T_{\text{SP}}(B). \end{cases} \quad (5)$$

The local spin-correlation proxy remains finite above the transition and is therefore not treated as an order parameter.

4 Numerical procedure

Three scan families are generated:

Family	Varied parameter	Fixed condition	Channel
Temperature lattice	T	$B = 0$	dynamic lattice
Field lattice	B	$T = 7$	dynamic lattice
Temperature C_1	T	$B = 0$	static C_1

Each run saves the rephasing real spectrum, raw complex response arrays, and text/JSON summaries. The Phase 2 analysis extracts finite-window integrals around the hybridized diagonal and cross resonances.

5 Acceptance criteria

The temperature lattice scan should show the lattice parameters $\delta(T, 0)$ and $g_Q^{\text{eff}}(T, 0)$ decreasing to zero at and above T_{SP} . The spectral observables do not need to be perfectly monotonic because peak positions, finite windows, and linewidth effects all contribute, but a clear regime change should be visible across the transition.

The field scan should suppress the lattice channel as $T_{\text{SP}}(B)$ is driven below the fixed temperature. The C_1 static scan is not expected to vanish at T_{SP} ; persistence in that control is evidence that a surviving feature is not unique to the dimerised phase.

6 Requested figures and data

Phase 2 should produce:

- a scan-metrics table with one row per run;
- trend plots for the temperature lattice, field lattice, and temperature C_1 static scans;
- a comparison between lattice-dynamic and C_1 -static temperature scans;
- a text validation outcome and this updated protocol.

7 Interpretation of possible failures

If the lattice scan does not show a clear change at the transition, the finite-window metric may be too insensitive, the scan may need a finer grid near T_{SP} , or the chosen g_Q may be too small. If the C_1 static scan also vanishes at the transition, then the proxy no longer represents a uniform spin-correlation control and the trend model should be revised.

8 Numerical results

The Step 5 scan generated 19 spectra using $\eta = 0.001$, a 90×90 frequency grid, $n_{\text{bosons}} = 3$, and $\omega_Q = 0.035$ eV. The finite-window half-width used for the aggregate metrics is 0.025 eV.

Temperature scan	δ	g_Q^{eff}	V_{BD}^{static}	A_{cross}
$T = 4$	0.008452	0.011952	0.021952	17.211013
$T = 7$	0.007071	0.010000	0.020000	18.158347
$T = 10$	0.005345	0.007559	0.017559	16.783502
$T = 13$	0.002673	0.003780	0.013780	9.779390
$T = 14$	0	0	0.010000	8.519103
$T = 16$	0	0	0.010000	8.519103
$T = 20$	0	0	0.010000	8.519103

The temperature lattice scan shows the expected collapse of the dynamic lattice channel at the transition. At $T = 7$, $\delta = 0.007071$ and $g_Q^{\text{eff}} = 0.010000$, with main cross-window area 18.158347. At $T = 14$ and above, $\delta = g_Q^{\text{eff}} = 0$, and the same metric reaches the lattice-off plateau 8.519103.

Field scan	$T_{\text{SP}}(B)$	δ	g_Q^{eff}	A_{cross}
$B = 0$	14.000	0.007071	0.010000	18.158347
$B = 5$	12.600	0.006667	0.009428	18.473411
$B = 10$	8.400	0.004082	0.005774	12.119501
$B = 13$	4.536	0	0	8.519103
$B = 15$	1.400	0	0	8.519103

The field scan reproduces the same logic at fixed $T = 7$. Once $T_{\text{SP}}(B)$ is driven below the fixed temperature, the dimerisation and dynamic coupling vanish. The finite-window cross area then reaches the same baseline as the high-temperature lattice-off regime.

C_1 static scan	C_1	g_Q^{eff}	V_{BD}^{static}	A_{cross}
$T = 4$	-0.400111	0	0.022144	18.022625
$T = 7$	-0.378667	0	0.021493	18.473889
$T = 10$	-0.362583	0	0.021005	18.682094
$T = 13$	-0.350074	0	0.020625	18.664855
$T = 14$	-0.346500	0	0.020517	18.622864
$T = 16$	-0.340067	0	0.020322	18.502526
$T = 20$	-0.329471	0	0.020000	18.179912

The C_1 static control does not collapse at T_{SP} . At $T = 20$, where $\delta = 0$, it still gives $A_{\text{cross}} = 18.179912$, much larger than the lattice-off value 8.519103.

9 Validation outcome

The Step 5 scan validates the trend-level interpretation. The dimerisation $\delta(T, 0)$ decreases monotonically across the temperature scan and vanishes at and above T_{SP} . The dynamic lattice coupling g_Q^{eff} follows the same trend. The high-temperature lattice endpoint therefore has both $\delta = 0$ and $g_Q^{\text{eff}} = 0$.

The field scan also passes the control. At fixed $T = 7$, the high-field points $B = 13$ and $B = 15$ suppress the spin-Peierls transition temperature below the fixed temperature, giving $\delta = 0$ and $g_Q^{\text{eff}} = 0$. The cross-window area reaches the same lattice-off value as the high-temperature scan.

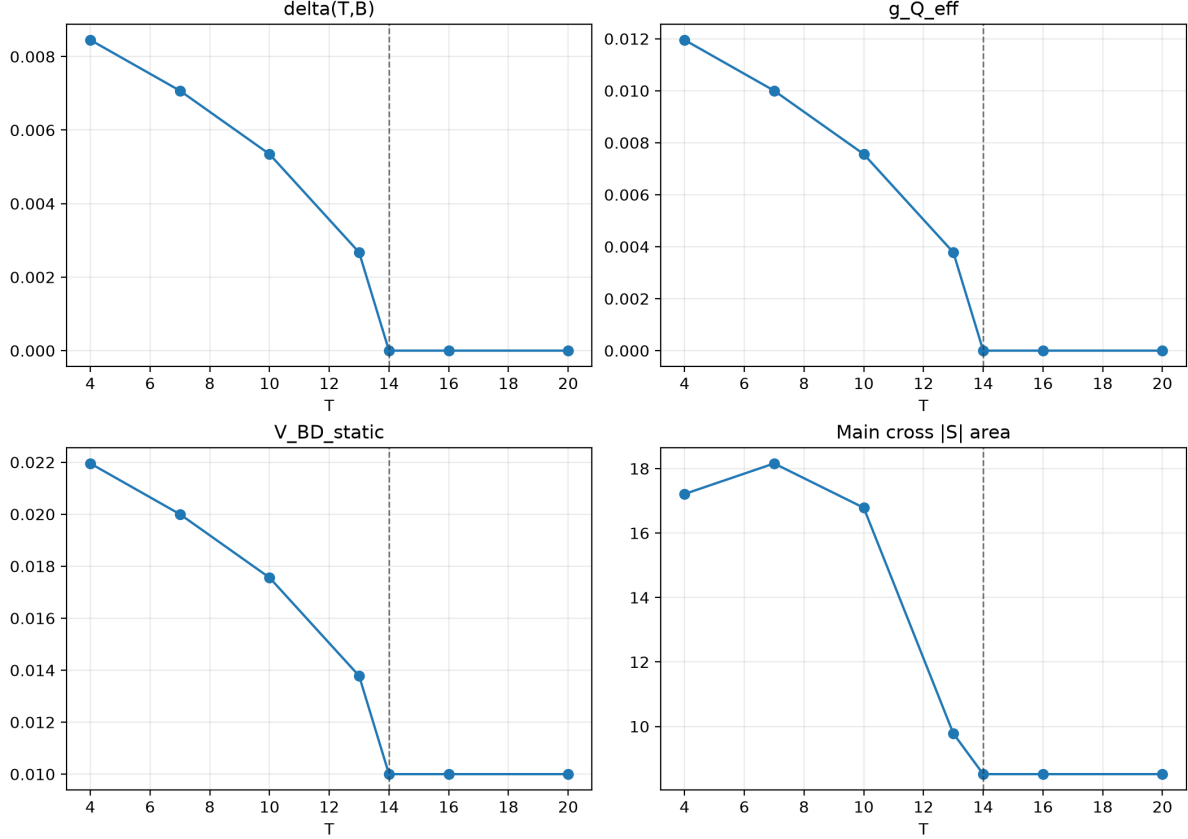


Figure 1: Temperature scan for the dynamic lattice channel. The vertical dashed line marks $T_{SP} = 14$.

The C_1 static scan provides the interpretive warning. It keeps finite C_1 above the transition and gives a strong cross-window signal even when $\delta = 0$. Therefore a nonlinear feature that persists above T_{SP} should not be assigned uniquely to the dimerised spin-Peierls phase. It may instead reflect a local spin-correlation contribution to bright-dark mixing.

10 Conclusion

Step 5 turns the discrete Step 4 controls into a practical trend diagnostic. For this minimal model, the lattice-dynamic signature follows the spin-Peierls order parameter and collapses to a common baseline when $\delta(T, B)$ vanishes. The C_1 static control remains large across the same temperature range. The next useful model step is therefore not another scalar trend control, but either a scan of the dynamical phonon parameters (g_Q, ω_Q) or a controlled reintroduction of momentum/spinon-sector structure.

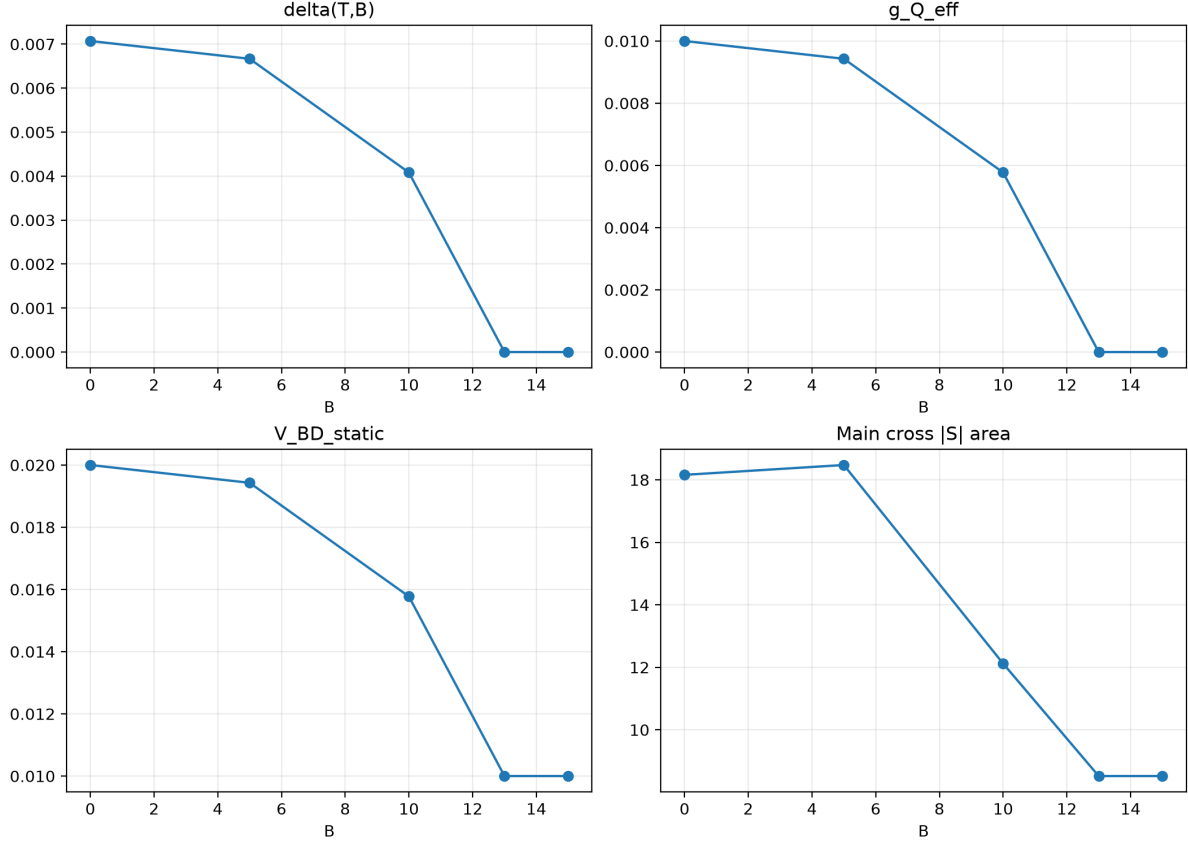


Figure 2: Field scan at fixed $T = 7$. The lattice channel is suppressed once $T_{SP}(B)$ falls below the fixed temperature.

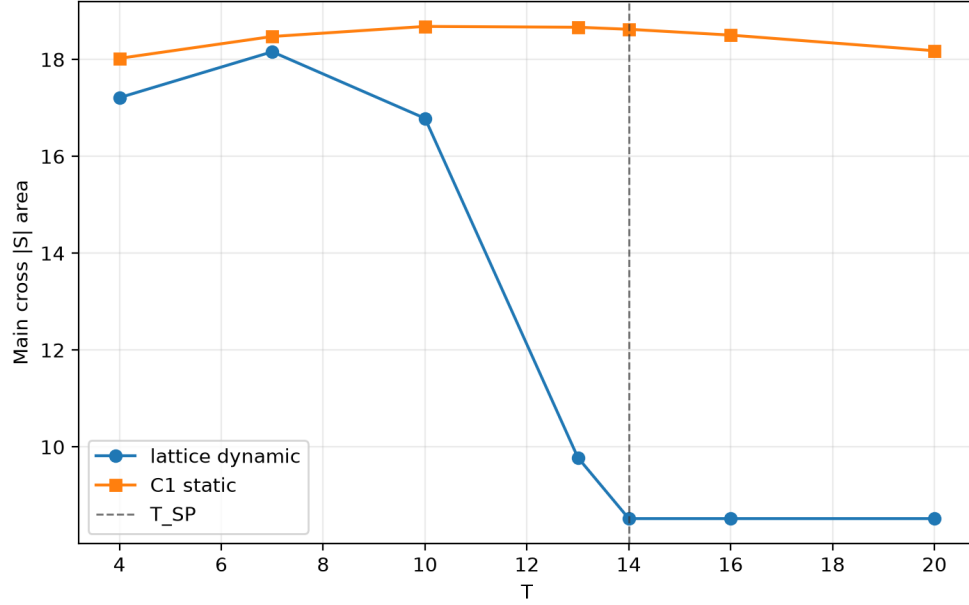


Figure 3: Temperature comparison between the dynamic lattice channel and the static C_1 control.